

NIPUNI DE SILVA

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LinkedIn : <https://www.linkedin.com/in/nipunidsr/>

PROFILE

A researcher specializing in machine learning, numerical modeling, and optimization, preparing my first first-author publication in a top-tier journal. I have presented my work at premier conferences including SIAM, SIAM-NNP, ICERM, and IMSI. I excel at transforming real-world data into insights on collective dynamics, developing algorithms, simulation frameworks, and data-driven inference models. In addition, I bring experience in project management, mentoring students, and cross-disciplinary collaboration.

EDUCATION

Clarkson University

August 2019 - Present

Doctor of Philosophy in Mathematics.

Advisor: Dr. James Greene.

Thesis Title: Dynamics on Collective Motion.

Clarkson University

August 2019 - May 2022

Master of Science in Mathematics.

GPA: 3.83 out of 4.0.

University of Kelaniya

March 2014 - March 2018

Bachelor of Science in Mathematics (Minor - Computer Science).

GPA: 3.73 out of 4.0.

INDUSTRY EXPERIENCE

- Mathematical Problems in Industry (MPI) June 2025
Focused on improving the quality of synthetic data for training models predicting disease progression.
Co-designed the N-N weighting scheme.
Vironix Health and Society for Industrial and Applied Mathematics (SIAM).
- Graduate Student Mathematical Modeling Camp (GSMMC) June 2025
Focused on SVD, randomized least squares, and applications in cancer data.
Society for Industrial and Applied Mathematics (SIAM).

PUBLICATIONS

- **Nipuni de Silva**, James Greene and Ming Zhong. "Learning environmental and collective dynamics" (*in progress*).

RESEARCH TALKS AND WORKSHOPS

- Dynamics and Learning of Collective Systems. Oct 2025
Mathematical Association of America (MAA) -Seaway : Fall Meeting 2025.
- Dynamics and Learning of Interacting Particle Systems. April 2024
Clarkson University Research and Project Showcase (RAPS).
Clarkson University.
- Dynamics and Learning for Stochastic Interacting Particle Systems. March 2024
Mathematics Conference and Competition of Northern New York (MCCNNY).

- Parkinson Disease Patient Profiles in Different Healthcare Setups: A Comparative Analysis. April 2020
Clarkson University Research and Project Showcase (RAPS).
Clarkson University.

POSTER PRESENTATIONS

- Learning Interactions in Collective Dynamics. March 2025
Emergent Behavior in Complex Systems of Interacting Agents.
Institute for Mathematical and Statistical Innovations (IMSI).
- Dynamics and Learning for Interacting Particle Systems May 2024
Interacting Particle Systems: Analysis, Control, Learning and Computation
The Institute for Computational and Experimental Research in Mathematics (ICERM).
- Data-Driven Stochastic Differential Equation Models for Emergent Behavior in Collective Dynamics November 2023
SIAM New York-New Jersey-Pennsylvania Section Conference (SIAM-NNP).
- Deep Learning the Dynamical Model for Evolution of Two-Dimensional Vortex Dynamics. June 2023
SIAM Conference on Applications of Dynamical Systems (DS23), Portland.

TEACHING EXPERIENCE

Teaching Assistant, Department of Mathematics, Clarkson University August 2019 - Present

Instructor:

- MA 41 - Co-Calculus I (Fall 2021).
- MA 41 - Co-Calculus I (Spring 2023).
- MA 41 - Co-Calculus I (Spring 2024).
- MA 42 - Co-Calculus II (Fall 2022).
- MA 42 - Co-Calculus II (Fall 2024).
- MA 41 - Co-Calculus I (Fall 2025).

Teaching Assistant:

- MA 131 - Calculus II for Engineering (Spring 2025).
- MA 232 - Elementary Ordinary Differential Equations (Fall 2023).
- MA 232 - Elementary Ordinary Differential Equations (Fall 2022).
- MA 131 - Calculus I for Engineering (Spring 2022).
- MA 231 - Calculus III (Spring 2021).
- MA 132 - Calculus II for Engineering (Fall 2020).
- MA 132 - Calculus II for Engineering (Spring 2020).
- MA 131 - Calculus I for Engineering (Fall 2019).

Instructor, Department of Mathematics, University of Kelaniya February 2018 - January 2019

Instructor:

- Methods using appropriate software (MATLAB), Scientific Computing I using appropriate software (MATLAB).
- Elementary Ordinary Differential Equations.
- Discrete Mathematics for Computing.

Teaching Assistant:

- Methods using appropriate software (MATLAB), Scientific Computing II using appropriate software (MATLAB).

SERVICE, OUTREACH & LEADERSHIP

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| • President, Sri Lankan Student Association.
Clarkson University. | 2024 - 2025 |
| • Treasurer, Graduate Student Association.
Clarkson University. | 2024 - 2025 |
| • Lead Committee member, Travel grants, Graduate Student Association.
Clarkson University. | 2024 - 2025 |
| • Lead Committee member, Financial Bylaw Revision, Graduate Student Association.
Clarkson University. | 2024 - 2025 |
| • Committee member, Association bylaw Revision, Graduate Student Association.
Clarkson University. | 2024 - 2025 |
| • Secretary, Sri Lankan Student Association.
Clarkson University. | 2023 - 2024 |
| • Councilor, Graduate Student Association.
Clarkson University. | 2023 - 2024 |
| • Committee member, Travel grants, Graduate Student Association.
Clarkson University. | 2023 - 2024 |
| • Committee member, Academic Integrity Committee, Graduate Student Association.
Clarkson University. | 2023 - 2024 |
| • A student volunteer, VIDYA Exhibition.
Department of Mathematics, University of Kelaniya. | 2019 |

HONORS/AWARDS/GRANTS

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| • Graduate Student Teaching Award.
Clarkson's annual Celebration and Recognition of Excellence Weekend (CREW). | 2025 |
| • IMSI Travel Support.
The Institute for Mathematical and Statistical Innovation (IMSI). | 2025 |
| • NNP-SIAM Travel Award.
Society for Industrial and Applied Mathematics. | 2024 |
| • ICERM Travel Support.
Interacting Particle Systems: Analysis, Control, Learning and Computation (ICERM). | 2024 |
| • CUGSA Graduate Student Travel Award.
Clarkson University. | 2024 |

- CUGSA Graduate Student Travel Award. 2023
Clarkson University.
- Dean's List. 2017-2018
Faculty of Science, University of Kelaniya.
- Dean's List. 2016-2017
Faculty of Science, University of Kelaniya.

PROFESSIONAL AFFILIATIONS

- Association of Women in Mathematics (AWM).
- Society for Mathematical Biology (SMB).
- Society for Industrial and Applied Mathematics (SIAM).
- Pi Mu Epsilon.

REFERENCES

Dr. James Greene, Assistant Professor, Clarkson University, Potsdam, NY, USA

Dr. Guangming Yao, Department Chair, Clarkson University, Potsdam, NY, USA

Dr. Thevasha Sathiyakumar, Visiting Assistant Professor, Clarkson University, Potsdam, NY, USA

Prof. Christopher Martin, Instructor, Clarkson University, Potsdam, NY, USA